ALICE L WALTON SCHOOL OF MEDICINE

Neural Control of the GI System

Last updated July 9, 2026 · 18 min read · CC BY-SA 4.0

Author: Ryan Downey
Contributors: Ian Murray, Eleanor Schrems, Kevin Kunkler, Jordan Finn, Russell Johnson

Listen to this Brick
0:00 / 33:07 1x

Learning Objectives (5)

After completing this brick, you will be able to:

1. Explain the structural organization of the enteric nervous system and functions of myenteric and submucosal plexuses.
2. Trace the short-loop reflex pathway through intrinsic sensory neurons, interneurons, and motor neurons in the ENS.
3. Predict the functional consequences of sympathetic vs. parasympathetic modulation of ENS activity on GI motility and secretion.
4. Explain the role of enteroendocrine cells as luminal sensory transducers that link chyme composition to ENS and CNS signaling.
5. Trace the neural and hormonal pathways by which cephalic stimuli initiate gastric secretory activity before food arrives in the stomach.

CASE CONNECTION

AM, a 52-year-old female, presents at the gastroenterology clinic with unexplained bloating and altered bowel patterns persisting for several weeks. She reports no recent dietary changes, no fever, and no significant weight loss.

What nervous system mechanisms could explain AM's bloating and altered bowel patterns? How might changes in nervous or endocrine signaling account for altered motility and secretion? Consider your answers as you read, and we'll revisit AM at the end of the brick.

GO TO CONCLUSION ↓

ENS Structural Organization: Myenteric & Submucosal Plexuses (LO1)

The enteric nervous system (ENS) is a semi-autonomous neural network embedded within the wall of the gastrointestinal (GI) tract. The ENS contains intrinsic neurons whose cell bodies lie within the gut wall and also receives input from extrinsic neurons with cell bodies in the CNS and autonomic ganglia. Often called [illegible] approximately 500 million neurons, more than the spinal cord, and can coordinate complex GI functions entirely independent of central nervous system (CNS) input. It extends from the esophagus to the anus and is organized into two primary ganglionated plexuses: the myenteric plexus and the submucosal plexus (SMP).

Myenteric Plexus Functions

The myenteric plexus (MP), also known as Auerbach's plexus, is located between the longitudinal and circular smooth muscle layers of the GI wall, a position that makes it the primary regulator of GI motility. Neurons of the MP coordinate the contraction and relaxation of smooth muscle to produce peristalsis—the propulsive wave that moves luminal contents aborally—as well as segmentation, which mixes chyme with digestive secretions.

The MP spans the entire length of the GI tract and is particularly dense in regions with high motility demands, such as the small intestine and colon. Disruption of MP function—whether through neuropathy, inflammation, or surgical damage—can result in dysmotility disorders such as gastroparesis or intestinal pseudo-obstruction.

CLINICAL CORRELATION

Hirschsprung's Disease: When Myenteric Ganglia Are Absent

Failure of neural crest cell migration leaves the distal colon — most commonly the rectosigmoid — without myenteric and submucosal ganglia. Without inhibitory motor neurons, the aganglionic segment cannot relax, producing functional obstruction and progressive proximal dilation (**megacolon**). The classic neonatal presentation is failur[illegible]th, accompanied by bilious vomiting and abdominal distension. Diagnosis is confirmed by rectal suction biopsy demonstrating absent ganglion cells, and treatment is surgical resection of the aganglionic segment.

Submucosal Plexus Functions

The submucosal plexus, also called Meissner's plexus, resides in the submucosal layer between the circular muscle and the mucosa. Unlike the MP, the SMP is primarily concerned with regulating mucosal secretion, epithelial transport, and local blood flow rather than motility. Neurons of the SMP innervate secretory glands, goblet cells, and enterocytes, modulating the release of water, electrolytes, and mucus into the gut lumen.

The SMP also receives sensory input from the mucosa and plays a key role in coordinating secretomotor reflexes—responses that couple luminal sensing to secretory output. For example, when luminal contents irritate the mucosa, SMP neurons can trigger increased fluid secretion to dilute and flush the irritant. This makes the SMP a critical interface between the luminal environment and the body's secretory defenses, with clinical relevance in conditions like secretory diarrhea and inflammatory bowel disease.

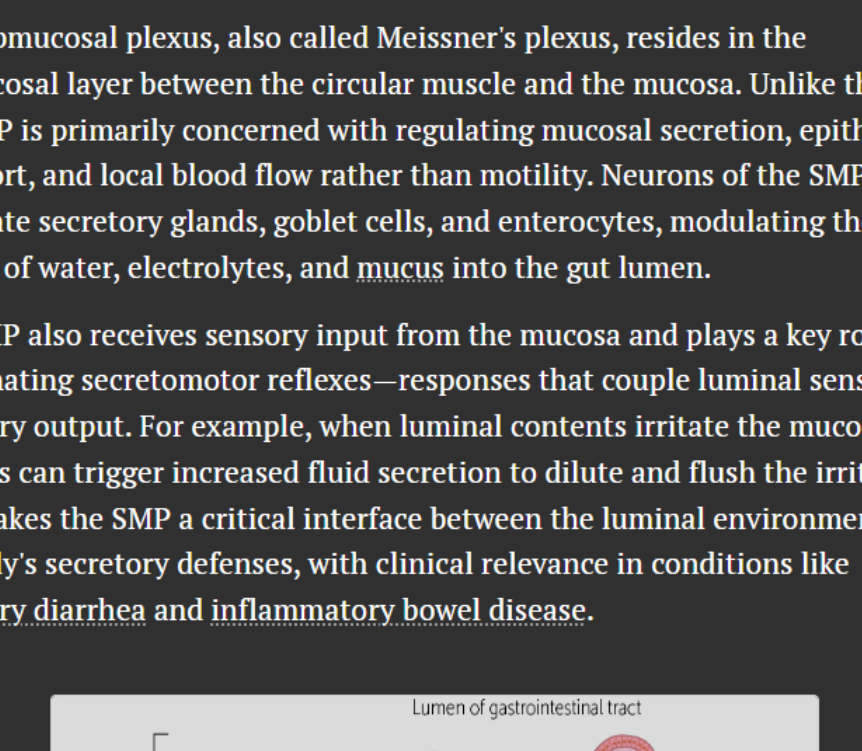

Figure 1: Organization of the gastrointestinal tract lumen.

What are the locations and primary functions of the myenteric versus submucosal plexus?

ENS Short-Loop Reflex Pathway (LO2)

The ENS operates through intrinsic reflex circuits that allow the gut to respond to luminal stimuli without requiring input from the brain or spinal cord. These short-loop reflex pathways are entirely contained within the ENS and involve three functional neuron types: intrinsic sensory neurons, interneurons, and motor neurons. Together, these neurons form a local arc that detects a stimulus, integrates the signal, and drives an appropriate effector response—all within milliseconds and without conscious awareness.

Intrinsic Sensory Neurons

Intrinsic primary afferent neurons (IPANs) are the first link in the ENS reflex arc. Located in both the MP and SMP, IPANs have sensory endings that project into the mucosa or respond to mechanical deformation of the gut wall. IPANs detect a variety of luminal stimuli, including mechanical distension, changes in luminal pH, osmolarity, and the presence of specific nutrients or irritants.

When activated, IPANs release neurotransmitters—most notably acetylcholine (ACh) and substance P—onto interneurons and motor neurons within the ENS. Luminal stimuli also activate enteroendocrine cells (EECs)—particularly enterochromaffin (EC) cells—which release serotonin (5-HT) basally onto IPAN terminals, amplifying sensory signaling within the reflex circuit.

Interneurons and Integration

Interneurons within the ENS serve as integrative hubs, receiving input from IPANs and distributing processed signals to motor neurons both orally (ascending) and aborally (descending) along the gut. Ascending interneurons typically use ACh as their primary neurotransmitter and project toward the oral end of the gut, where [illegible]ns to produce contraction. Descending interneurons, which use a combination of ACh, nitric oxide (NO), and vasoactive intestinal peptide (VIP), project aborally to activate inhibitory motor neurons, producing relaxation ahead of the bolus.

This ascending excitation and descending inhibition forms the neural basis of the peristaltic reflex—a coordinated, directional wave of contraction and relaxation that propels luminal contents through the GI tract. The interneuron network also allows for spatial and temporal summation of signals, enabling the ENS to generate graded responses calibrated to the nature and intensity of the stimulus.

Motor Neuron Output

Motor neurons are the final effectors of the ENS reflex arc, translating integrated neural signals into physiological responses. Excitatory motor neurons release ACh and substance P onto smooth muscle cells, causing contraction. Inhibitory motor neurons release NO, VIP, and ATP, causing smooth muscle relaxation. The balance between excitatory and inhibitory motor output determines the net contractile state of any given segment of the gut.

Beyond motility, motor neurons also innervate secretory cells in the mucosa and submucosa. Secretomotor neurons are a subset of motor neurons in the SMP that stimulate epithelial secretion of chloride and water, contributing to the fluid environment of the gut lumen. This dual role, governing both smooth muscle contraction and epithelial secretion, reflects the ENS's capacity for autonomous, integrated gut regulation.

CLINICAL CORRELATION

IBS and Altered ENS / EC Cell Signaling

Irritable bowel syndrome involves altered ENS sensitivity rather than structural disease. In affected patients, excess 5-HT release from EC cells or reduced SERT-mediated reuptake lowers the activation threshold of intrinsic primary afferent neurons (IPANs), generating exaggerated peristaltic and secretomotor reflexes. The dominant motor outputs determine the clinical subtype: diarrhea-predominant (IBS-D), constipation-predominant (IBS-C), or mixed (IBS-M). Treatment targets the same serotonergic machinery — 5-HT$_3$ antagonists such as **alosetron** reduce IPAN activation in IBS-D, while 5-HT$_4$ agonists enhance propulsive activity in IBS-C and chronic constipation.

ACTIVE LEARNING QUESTION

A bolus of food distends the small intestine wall. Which sequence correctly describes the ENS short-loop reflex response?

- **A.** ○ Motor neuron → IPAN → interneuron
- **B.** ○ IPAN → motor neuron directly
- **C.** ○ IPAN → interneuron → motor neuron
- **D.** ○ Interneuron → IPAN → motor neuron
- **E.** ○ CNS → spinal cord → motor neuron

Explanation (requires correct answer)

Sympathetic vs. Parasympathetic Modulation of ENS Activity (LO3)

Although the ENS can function autonomously, it is continuously modulated by the autonomic nervous system (ANS). Sympathetic and parasympathetic inputs do not directly control GI smooth muscle in most cases; instead, they act primarily by modulating ENS activity, enhancing or suppressing intrinsic reflex circuits to coordinate GI function with systemic physiological states such as the fight-or-flight and rest-and-digest responses.

Sympathetic Modulation

Sympathetic innervation of the GI tract originates from preganglionic neurons in the thoracolumbar spinal cord (T5–L2), which synapse in prevertebral ganglia (celiac, superior mesenteric, and inferior mesenteric ganglia). Postganglionic sympathetic fibers then project to the ENS, where they release norepinephrine (NE) onto ENS neurons and, to a lesser extent, directly onto smooth muscle and secretory cells.

The net effect of sympathetic activation on the GI tract is inhibitory: motility decreases, secretion is reduced, and sphincters contract. NE acts primarily on α2-adrenergic receptors on ENS neurons, suppressing the release of excitatory neurotransmitters like ACh. This "braking" effect on the ENS is physiologically appropriate during s[illegible]redirected away from the gut. A clinically relevant parallel: opioid analgesics cause constipation by suppressing excitatory ENS output via μ-opioid receptors on myenteric neurons—a mechanism distinct from adrenergic signaling but producing the same net inhibition of propulsive motility.

Parasympathetic Modulation

Parasympathetic innervation of the GI tract is primarily delivered via the vagus nerve (cranial nerve X), which supplies the esophagus, stomach, small intestine, and proximal colon. The distal colon, rectum, and internal anal sphincter receive parasympathetic input from the pelvic splanchnic nerves (S2–S4). Preganglionic parasympathetic fibers synapse directly on ENS neurons within the gut wall, where they release ACh onto nicotinic receptors.

Parasympathetic activation has a broadly excitatory effect on GI function: it increases motility, stimulates secretion, and relaxes sphincters. By activating ENS interneurons and motor neurons, parasympathetic input amplifies the intrinsic ENS circuits rather than bypassing them. This is why vagotomy—surgical cutting of the vagus nerve—reduces gastric acid secretion and motility, a procedure historically used to treat peptic ulcer disease. Anticholinergic medications, which block muscarinic receptors, reduce GI motility and secretion through this same pathway.

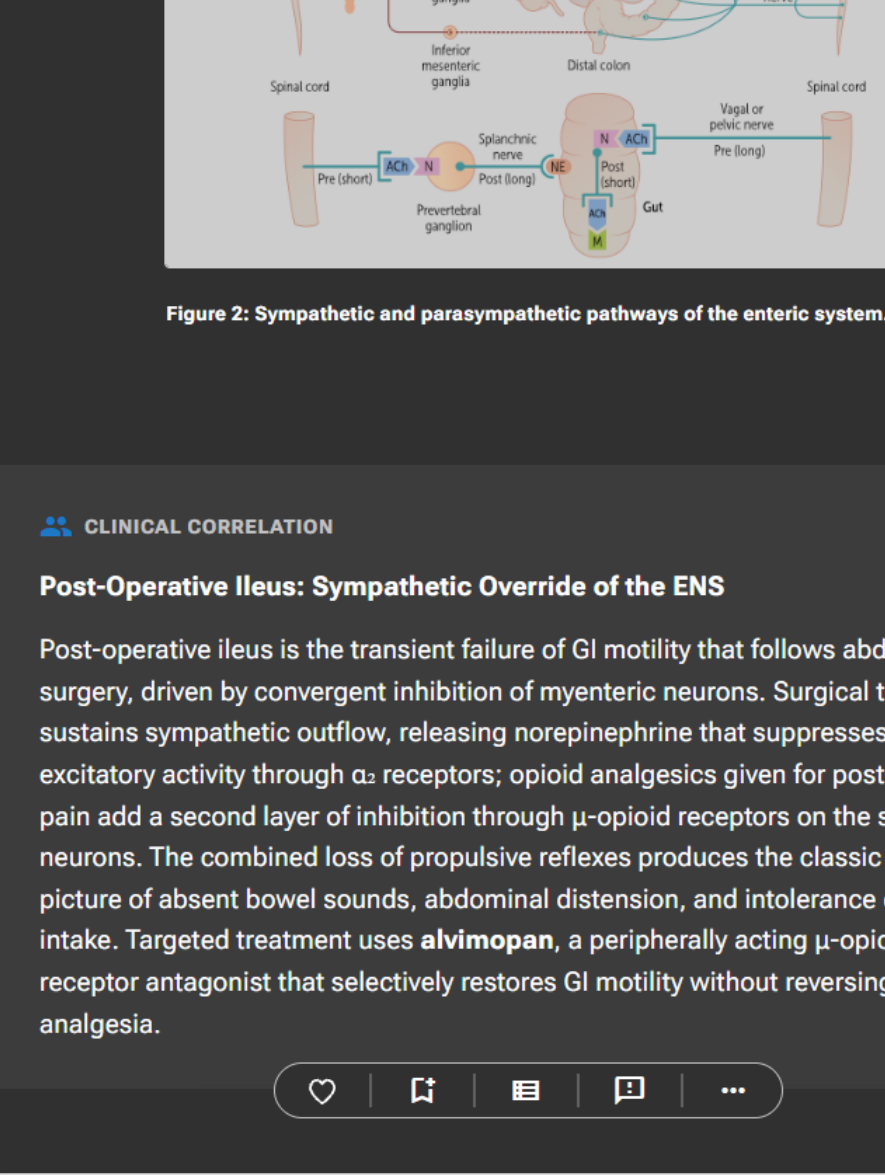

Figure 2: Sympathetic and parasympathetic pathways of the enteric system.

CLINICAL CORRELATION

Post-Operative Ileus: Sympathetic Override of the ENS

Post-operative ileus is the transient failure of GI motility that follows abdominal surgery, driven by convergent inhibition of myenteric neurons. Surgical trauma sustains sympathetic outflow, releasing norepinephrine that suppresses myenteric excitatory activity through α$_2$ receptors; opioid analgesics given for post-operative pain add a second layer of inhibition through μ-opioid receptors on the same neurons. The combined loss of propulsive reflexes produces the classic clinical picture of absent bowel sounds, abdominal distension, and intolerance of oral intake. Targeted treatment uses **alvimopan**, a peripherally acting μ-opioid receptor antagonist that selectively restores GI motility without reversing systemic analgesia.

ACTIVE LEARNING QUESTION

A patient receives a high-dose sympathomimetic agent during a medical emergency. Which GI effect is most expected?

- **A.** ○ Increased peristalsis
- **B.** ○ Decreased motility and secretion
- **C.** ○ Sphincter relaxation
- **D.** ○ Increased gastric acid output
- **E.** ○ Accelerated gastric emptying

Explanation (requires correct answer)

Enteroendocrine Cells as Luminal Sensory Transducers (LO4)

Enteroendocrine cells (EECs) are specialized epithelial cells scattered throughout the GI mucosa that function as luminal sensory transducers. Although they constitute less than 1% of intestinal epithelial cells, EECs collectively form the [illegible]positioned at the interface between the gut lumen and the underlying neural and vascular networks, EECs translate chyme composition into neural and hormonal signals that coordinate digestion.

Sensing Chyme Composition

EECs express chemosensory GPCRs on their apical (luminal) surface, for fatty acids, amino acids, sugars, bile acids, and short-chain fatty acids, enabling detection of the chemical properties of chyme. Distinct EEC subtypes respond to distinct nutrient classes with subtype-specific hormone release, coordinating the intestinal phase of digestion.

Enterochromaffin (EC) cells, the most abundant EEC subtype, are particularly important for ENS signaling. EC cells detect mechanical distension and chemical stimuli and respond by releasing serotonin (5-HT) from their basolateral surface. This 5-HT acts on IPANs to initiate peristaltic and secretomotor reflexes.

Neurotransmitter and Hormone Release

EECs communicate with the ENS and CNS through two primary mechanisms: paracrine signaling and endocrine signaling. In paracrine signaling, EECs release mediators—such as 5-HT, CCK, and peptide YY (PYY)—that diffuse locally to activate nearby ENS neurons, particularly IPANs and secretomotor neurons. This enables immediate ENS reflex responses without systemic circulation.

In endocrine signaling, EEC-derived hormones enter the bloodstream and act on distant targets, including the stomach, pancreas, liver, and CNS. For example, CCK release[illegible]stimulate pancreatic enzyme secretion and gallbladder contraction, while also activating vagal afferent fibers to signal satiety to the hypothalamus. This dual paracrine-endocrine function makes EECs essential integrators of digestive physiology, linking luminal nutrient sensing to coordinated systemic responses.

EEC mediators, particularly 5-HT and CCK, also activate vagal afferent terminals in the lamina propria, transmitting information about luminal composition to the nucleus tractus solitarius (NTS) in the brainstem. This gut-to-brain signaling pathway contributes to satiety, nausea, and reflexive modulation of gastric motility and secretion.

CLINICAL CORRELATION

Carcinoid Syndrome: EC Cell Overproduction of Serotonin

Carcinoid tumors arise from EC cells and secrete excess 5-HT and vasoactive peptides. Under normal physiology, hepatic first-pass metabolism clears these mediators from portal blood, limiting systemic exposure. When carcinoid tumors metastasize to the liver, this clearance is bypassed and the mediators reach the systemic circulation, producing the classic triad of episodic flushing, secretory diarrhea, and right-sided valvular fibrosis (tricuspid regurgitation and pulmonic stenosis). Diagnosis relies on elevated **24-hour urinary 5-HIAA**, the primary 5-HT metabolite. Treatment uses somatostatin analogues such as **octreotide** to suppress hormone release.

How do enteroendocrine cells function as luminal sensory transducers linking chyme to ENS and CNS?

Cephalic Phase of Gastric Secretion (LO5)

The cephalic phase of gastric secretion begins before food ever reaches the stomach. Sensory stimuli associated with food—its sight, smell, taste, and even the thought of eating—activate higher brain centers that initiate preparatory gastric secretory responses. This feedforward (anticipatory) phase accounts for approximately 20–30% of total gastric acid secretion during a meal and primes the stomach for efficient digestion. The cephalic phase is primarily neural — driven by vagal pathways to the gastric mucosa — with hormonal amplification downstream of vagal activation.

Neural Pathways

Cephalic stimuli are processed in the cerebral cortex, hypothalamus, and brainstem, where they activate the dorsal motor nucleus of the vagus (DMV). The DMV sends preganglionic parasympathetic fibers via the vagus nerve to the ENS of the stoma[illegible]s synapse on ENS neurons, which in turn stimulate parietal cells to secrete hydrochloric acid (HCl) and chief cells to secrete pepsinogen. Vagal stimulation also activates G cells in the antrum to release gastrin, amplifying acid secretion.

ENS neurons activated by vagal input release gastrin-releasing peptide (GRP) onto G cells in the gastric antrum, providing a potent stimulus for gastrin secretion. This neuro-hormonal amplification loop — vagal activation → ENS GRP release → gastrin secretion → parietal cell stimulation — illustrates how neural and hormonal signals are tightly integrated even in the earliest phase of digestion. Sham feeding experiments, in which subjects chew and spit food without swallowing, have confirmed that cephalic-phase acid secretion is entirely vagally mediated, as it is abolished by vagotomy.

Hormonal Amplification

The gastrin released by vagal-GRP stimulation (described above) acts on cholecystokinin-B (CCK-B) receptors on parietal cells to stimulate HCl secretion and on enterochromaffin-like (ECL) cells to release histamine, which further amplifies acid output via H$_2$ receptors on parietal cells. This cascade — vagal activation → ENS GRP → gastrin → ECL histamine → HCl — amplifies acid output in anticipation of a meal.

ENS motor neurons activated by vagal input also release ACh onto muscarinic (M3) receptors on parietal cells, providing a direct neural stimulus for acid secretion alongside the gastrin-mediated pathway. Vagal input also drives mucus secretion from mucous cells, protecting the gastric lining from the acidic environment. Clinically, proton pump inhibitors (PPIs) and H2 receptor antagonists target downstream components of this pathway, effectively reducing acid secretion regardless of whether the stimulus is cephalic, gastric, or intestinal.

ACTIVE LEARNING QUESTION

Which pathway best explains gastric acid secretion triggered by the smell of food before eating?

- **A.** ○ Sympathetic → parietal cell
- **B.** ○ Vagus → ENS → parietal cell
- **C.** ○ Gastrin alone, no neural input
- **D.** ○ CCK from duodenum
- **E.** ○ Secretin from S cells

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AM, what ENS-related mechanisms could explain her bloating and altered bowel patterns? How might disrupted communication between nervous and endocrine signals affect GI motility and secretion?

AM's symptoms like[illegible]the ENS. Altered myenteric plexus activity could impair peristalsis, leading to dysmotility and bloating. Disrupted submucosal plexus signaling may affect secretion and fluid balance, contributing to altered stool consistency. Changes in EEC function—such as abnormal 5-HT release from enterochromaffin cells—could dysregulate both motility reflexes and visceral sensation. Autonomic imbalance, such as excess sympathetic tone, would further suppress ENS activity. The integration of ENS structure, reflex pathways, autonomic regulation, and enteroendocrine signaling provides insight into how GI disorders arise and guides clinical diagnostic reasoning.

Summary

- The ENS contains two main plexuses: the myenteric plexus (Auerbach's), located between muscle layers and controlling motility, and the submucosal plexus (Meissner's), located in the submucosa and controlling secretion and local blood flow.
- ENS short-loop reflexes operate through a three-neuron arc: IPANs detect luminal stimuli, interneurons integrate and distribute signals orally and aborally, and motor neurons drive smooth muscle contraction/relaxation and mucosal secretion.
- Sympathetic input (via norepinephrine on α2 receptors) inhibits ENS activity, reducing motility and secretion; parasympathetic input (via vagal ACh) excites ENS circuits, increasing motility and secretion.
- Enteroendocrine cells act as luminal sensory transducers, detecting nutrients via apical GPCRs and releasing mediators (e.g., 5-HT, CCK, secretin) basally to activate ENS neurons and systemic hormonal pathways.
- The cephalic phase of gastric secretion is initiated by sensory stimuli (sight, smell, taste of food) activating the vagus nerve, which stimulates ENS neurons, parietal cells, and G cells to release gastrin, driving preparatory HCl secretion.
- The vagal-GRP-gastrin-histamine cascade amplifies cephalic-phase acid secretion, accounting for 20–30% of total meal-stimulated acid output; this pathway is targeted clinically by PPIs and H2 receptor antagonists.

Review Questions

1. What is the primary function of the myenteric plexus in the GI tract?

- **A.** ○ Controls GI motility
- **B.** ○ Regulates secretion and blood flow
- **C.** ○ Mediates nutrient absorption
- **D.** ○ Detects luminal sensory stimuli
- **E.** ○ Releases gut hormones

Explanation (requires correct answer)

2. Which sequence correctly describes the ENS short-loop reflex after intestinal distension?

- **A.** ○ Motor neuron → IPAN → interneuron
- **B.** ○ IPAN → motor neuron directly
- **C.** ○ IPAN → interneuron → motor neuron
- **D.** ○ Interneuron → IPAN → motor neuron
- **E.** ○ CNS → spinal cord → motor neuron

Explanation (requires correct answer)

3. How does sympathetic activity modulate ENS function in the GI tract?

- **A.** ○ Increases peristalsis
- **B.** ○ Decreases motility and secretion
- **C.** ○ Causes sphincter relaxation
- **D.** ○ Increases gastric acid output
- **E.** ○ Accelerates gastric emptying

Explanation (requires correct answer)

4. What role do enteroendocrine cells play in gut physiology?

- **A.** ○ Drive smooth muscle contraction
- **B.** ○ Detect nutrients and release hormones
- **C.** ○ Send direct electrical signals to CNS
- **D.** ○ Act as primary gut pacemakers
- **E.** ○ Constitute most epithelial cells

Explanation (requires correct answer)

5. Which pathway initiates gastric acid secretion during the cephalic phase?

- **A.** ○ Sympathetic → parietal cell
- **B.** ○ Vagus → ENS → parietal cell
- **C.** ○ Gastrin alone, no nerves
- **D.** ○ CCK from duodenum
- **E.** ○ Secretin from S cells

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ Explain the structural organization of the enteric nervous system and functions of myenteric and submucosal plexuses.
- 02 ☐ Trace the short-loop reflex pathway through intrinsic sensory neurons, interneurons, and motor neurons in the ENS.
- 03 ☐ Predict the functional consequences of sympathetic vs. parasympathetic modulation of ENS activity on GI motility and secretion.
- 04 ☐ Explain the role of enteroendocrine cells as luminal sensory transducers that link chyme composition to ENS and CNS signaling.
- 05 ☐ Trace the neural and hormonal pathways by which cephalic stimuli initiate gastric secretory activity before food arrives in the stomach.

Objective Confidence: 0/5

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ Explain the structural organization of the enteric nervous system and functions of myenteric and submucosal plexuses.
- 02 ☐ Trace the short-loop reflex pathway through intrinsic sensory neurons, interneurons, and motor neurons in the ENS.
- 03 ☐ Predict the functional consequences of sympathetic vs. parasympathetic modulation of ENS activity on GI motility and secretion.
- 04 ☐ Explain the role of enteroendocrine cells as luminal sensory transducers that link chyme composition to ENS and CNS signaling.
- 05 ☐ Trace the neural and hormonal pathways by which cephalic stimuli initiate gastric secretory activity before food arrives in the stomach.

Objective Confidence: 0/5